

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Carbon monoxide (co)
Observed / expected baseline	0.5 / 0.136957 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-29 17:00 UTC
Location	29.7337, -95.2576
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 865	40.37 to 100; mean 72.9254	59.12 at 2026-07-29 16:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 865	22.7778 to 36.1111; mean 29.651	32 at 2026-07-29 16:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 1160	0 to 270; mean 163.543	190 at 2026-07-29 16:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 1212	0 to 7.71666; mean 2.8498	2.05778 at 2026-07-29 16:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	10 / 120	5909.24 to 5941.31; mean 5927.82	5928.39 at 2026-07-29 18:00Z; 1 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	10 / 120	48.9097 to 2294.53; mean 860.215	2188.77 at 2026-07-29 18:00Z; 1 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	10 / 120	34.4259 to 49.1773; mean 40.4243	43.2396 at 2026-07-29 18:00Z; 1 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	10 / 120	1004.76 to 1014.21; mean 1009.87	1010.7 at 2026-07-29 18:00Z; 1 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	10 / 120	18.3487 to 22.1724; mean 20.4638	19.1542 at 2026-07-29 18:00Z; 1 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	10 / 120	-3.50088 to 2.79912; mean 0.6582	1.74827 at 2026-07-29 18:00Z; 1 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	10 / 120	0.334747 to 6.23475; mean 3.0269	3.18474 at 2026-07-29 18:00Z; 1 h after event
openaq	Ozone (ozone)	ppm	17 / 1158	-0.004 to 0.053; mean 0.0177	0.018 at 2026-07-29 17:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 216	16 to 143; mean 41.4167	120 at 2026-07-29 17:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	69 / 2909	0 to 37.5; mean 6.2519	15.7 at 2026-07-29 17:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 39.7528	33 at 2026-07-29 17:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 360	49 to 93; mean 74.7111	69 at 2026-07-29 17:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 0; mean 0	0 at 2026-07-29 17:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 360	1009 to 1016; mean 1012.32	1013 at 2026-07-29 17:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 360	24.98 to 36.85; mean 30.463	33.72 at 2026-07-29 17:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 321; mean 192.231	208 at 2026-07-29 17:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.45 to 6.55; mean 2.2293	2.42 at 2026-07-29 17:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	74 / 5222	0 to 232; mean 8.0665	8.3 at 2026-07-29 17:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-07-29 18:13Z; 1 h 14 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-07-29 18:13Z; 1 h 14 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	9 / 9	0.0256055 to 0.0278855; mean 0.0271	0.0256055 at 2026-07-29 18:13Z; 1 h 14 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-07-29 18:13Z; 1 h 14 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	7 / 7	0.000175431 to 0.0002266; mean 0.0002	0.000223665 at 2026-07-29 18:13Z; 1 h 14 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-29 18:13Z; 1 h 14 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	5 / 5	3.65242e-05 to 4.17212e-05; mean 0	4.17212e-05 at 2026-07-29 18:44Z; 1 h 45 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-29 18:44Z; 1 h 45 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-07-29 18:13Z; 1 h 14 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	7 / 7	-7.46696e-06 to 0.000148654; mean 0.0001	0.000148654 at 2026-07-29 18:13Z; 1 h 14 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-29 18:13Z; 1 h 14 min after event
tceq	Carbon monoxide (co)	ppm	3 / 188	0.1 to 0.8; mean 0.2069	0.5 at 2026-07-29 17:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 707	-2.6 to 19.7; mean 4.2532	10.5 at 2026-07-29 17:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 190	-0.7 to 5.7; mean -0.0842	-0.1 at 2026-07-29 17:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-28 00Z 85.67, 06Z 89.53, 12Z 71.24, 18Z 55.87, 07-29 00Z 76.33, 06Z 90.25, 12Z 71.09, 18Z 51.84, 07-30 00Z 78.78, 06Z 91.49, 12Z 70.66, 18Z 50.99, 07-31 00Z 75.4
asos	Air temperature (temperature)	07-28 00Z 26.3, 06Z 25.4, 12Z 30.09, 18Z 33.68, 07-29 00Z 28.38, 06Z 25.36, 12Z 30.45, 18Z 34.3, 07-30 00Z 28.79, 06Z 26.07, 12Z 30.35, 18Z 34.32, 07-31 00Z 28.94
asos	Wind direction (wind_direction)	07-28 00Z 153.8, 06Z 118.1, 12Z 203.8, 18Z 186.3, 07-29 00Z 173, 06Z 85.91, 12Z 191.5, 18Z 193.3, 07-30 00Z 165.2, 06Z 110.7, 12Z 174.6, 18Z 182.7, 07-31 00Z 175.2
asos	Wind speed (wind_speed)	07-28 00Z 2.209, 06Z 1.343, 12Z 2.956, 18Z 4.302, 07-29 00Z 3.118, 06Z 0.9042, 12Z 3.001, 18Z 4.181, 07-30 00Z 3.425, 06Z 1.402, 12Z 2.307, 18Z 3.84, 07-31 00Z 3.595
noaa_gfs	500-hPa geopotential height (gh_500)	07-28 06Z 5934, 12Z 5917, 18Z 5931, 07-29 00Z 5923, 06Z 5927, 12Z 5911, 18Z 5929, 07-30 00Z 5924, 06Z 5933, 12Z 5924, 18Z 5940, 07-31 00Z 5940
noaa_gfs	Planetary boundary layer height (pbl_height)	07-28 06Z 357.2, 12Z 278.8, 18Z 1728, 07-29 00Z 1092, 06Z 330.9, 12Z 239, 18Z 2096, 07-30 00Z 1062, 06Z 384.3, 12Z 235.3, 18Z 1652, 07-31 00Z 866.8
noaa_gfs	Precipitable water (precipitable_water)	07-28 06Z 35.49, 12Z 37.92, 18Z 41.7, 07-29 00Z 42.79, 06Z 36.52, 12Z 39.26, 18Z 43.56, 07-30 00Z 46, 06Z 36.57, 12Z 38.68, 18Z 41.87, 07-31 00Z 44.75
noaa_gfs	Surface pressure (surface_pressure)	07-28 06Z 1010, 12Z 1010, 18Z 1010, 07-29 00Z 1008, 06Z 1009, 12Z 1010, 18Z 1010, 07-30 00Z 1008, 06Z 1010, 12Z 1012, 18Z 1012, 07-31 00Z 1010
noaa_gfs	850-hPa temperature (t_850)	07-28 06Z 21.47, 12Z 20.75, 18Z 18.75, 07-29 00Z 20.13, 06Z 21.27, 12Z 20.59, 18Z 19.02, 07-30 00Z 20.47, 06Z 21.66, 12Z 21.17, 18Z 19.25, 07-31 00Z 21.05
noaa_gfs	10-m eastward wind (u_10m)	07-28 06Z 0.8741, 12Z 1.111, 18Z 0.9095, 07-29 00Z -0.3137, 06Z 1.144, 12Z 0.9747, 18Z 1.498, 07-30 00Z 0.2491, 06Z 0.8699, 12Z 1.155, 18Z 0.5909, 07-31 00Z -1.165
noaa_gfs	10-m northward wind (v_10m)	07-28 06Z 3.028, 12Z 1.826, 18Z 2.723, 07-29 00Z 4.9, 06Z 2.639, 12Z 1.856, 18Z 2.865, 07-30 00Z 4.345, 06Z 3.252, 12Z 1.391, 18Z 2.493, 07-31 00Z 5.004
openaq	Ozone (ozone)	07-28 00Z 0.01687, 06Z 0.00992, 12Z 0.01452, 18Z 0.02731, 07-29 00Z 0.0169, 06Z 0.008722, 12Z 0.01348, 18Z 0.02853, 07-30 00Z 0.01801, 06Z 0.01094, 12Z 0.01402, 18Z 0.03224, 07-31 00Z 0.01692
openaq	PM10 (pm10)	07-28 00Z 18.67, 06Z 25.28, 12Z 64.11, 18Z 58.17, 07-29 00Z 46.94, 06Z 37.56, 12Z 58.22, 18Z 45.39, 07-30 00Z 26.22, 06Z 22.17, 12Z 51.61, 18Z 37.73, 07-31 00Z 26.78
openaq	PM2.5 (pm25)	07-28 00Z 5.887, 06Z 5.726, 12Z 5.828, 18Z 5.582, 07-29 00Z 6.283, 06Z 5.394, 12Z 6.89, 18Z 6.695, 07-30 00Z 7.032, 06Z 7.505, 12Z 7.364, 18Z 5.105, 07-31 00Z 5.069
openweather	Cloud cover (cloud_cover)	07-28 00Z 11.6, 06Z 1.833, 12Z 1.552, 18Z 0.7333, 07-29 00Z 9.161, 06Z 1.367, 12Z 13.83, 18Z 81.55, 07-30 00Z 52.9, 06Z 85.37, 12Z 72.52, 18Z 79.32, 07-31 00Z 86.68
openweather	Relative humidity (humidity)	07-28 00Z 80.2, 06Z 87.7, 12Z 78.45, 18Z 60.73, 07-29 00Z 73.84, 06Z 88.6, 12Z 78.34, 18Z 58.29, 07-30 00Z 74.53, 06Z 88.77, 12Z 78.69, 18Z 56.52, 07-31 00Z 72.32
openweather	Precipitation rate (precipitation)	07-28 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-29 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-30 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-31 00Z 0
openweather	Surface pressure (pressure)	07-28 00Z 1012, 06Z 1012, 12Z 1013, 18Z 1011, 07-29 00Z 1010, 06Z 1012, 12Z 1013, 18Z 1011, 07-30 00Z 1012, 06Z 1013, 12Z 1015, 18Z 1013, 07-31 00Z 1014
openweather	Air temperature (temperature)	07-28 00Z 27.48, 06Z 26.09, 12Z 29.53, 18Z 35.12, 07-29 00Z 29.97, 06Z 26.18, 12Z 29.85, 18Z 35.49, 07-30 00Z 30.36, 06Z 26.84, 12Z 29.83, 18Z 35.77, 07-31 00Z 30.64
openweather	Wind direction (wind_direction)	07-28 00Z 193.6, 06Z 197.4, 12Z 212.3, 18Z 190.2, 07-29 00Z 180.2, 06Z 162.1, 12Z 217.1, 18Z 192.4, 07-30 00Z 175.4, 06Z 186.9, 12Z 233, 18Z 188.2, 07-31 00Z 171.3
openweather	Wind speed (wind_speed)	07-28 00Z 3.204, 06Z 2.142, 12Z 2.101, 18Z 2.27, 07-29 00Z 2.553, 06Z 1.714, 12Z 2.068, 18Z 2.284, 07-30 00Z 2.733, 06Z 2.038, 12Z 1.981, 18Z 2.143, 07-31 00Z 2.596

Source	Metric	Values
purpleair	PM2.5 (pm25)	07-28 00Z 7.9, 06Z 7.894, 12Z 6.575, 18Z 6.549, 07-29 00Z 8.031, 06Z 7.668, 12Z 8.763, 18Z 8.366, 07-30 00Z 9.735, 06Z 10.2, 12Z 9.022, 18Z 6.542, 07-31 00Z 7.317
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	CO column (s5p_co_column)	07-28 18Z 0.02775, 07-29 18Z 0.02655, 07-30 18Z 0.02743
sentinel5p	CO granules available (s5p_co_granule_available)	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-28 18Z 0.0001861, 07-29 18Z 0.0002022, 07-30 18Z 0.0002134
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-28 18Z 3.894e-05, 07-29 18Z 3.912e-05, 07-30 18Z 3.936e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-28 18Z 4.926e-05, 07-29 18Z 6.457e-05, 07-30 18Z 2.458e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
tceq	Carbon monoxide (co)	07-28 06Z 0.1944, 12Z 0.2294, 18Z 0.1778, 07-29 00Z 0.16, 06Z 0.2111, 12Z 0.2722, 18Z 0.1944, 07-30 00Z 0.21, 06Z 0.2, 12Z 0.2313, 18Z 0.1944, 07-31 00Z 0.1778
tceq	Nitrogen dioxide (no2)	07-28 06Z 4.89, 12Z 5.902, 18Z 3.188, 07-29 00Z 2.288, 06Z 5.693, 12Z 5.49, 18Z 3.479, 07-30 00Z 2.292, 06Z 3.799, 12Z 5.327, 18Z 3.942, 07-31 00Z 2.124
tceq	Sulfur dioxide (so2)	07-28 06Z -0.2111, 12Z 0.08333, 18Z -0.1556, 07-29 00Z -0.09, 06Z -0.2, 12Z -0.1722, 18Z -0.1333, 07-30 00Z -0.1222, 06Z -0.1889, 12Z 0.4278, 18Z -0.1556, 07-31 00Z -0.1444

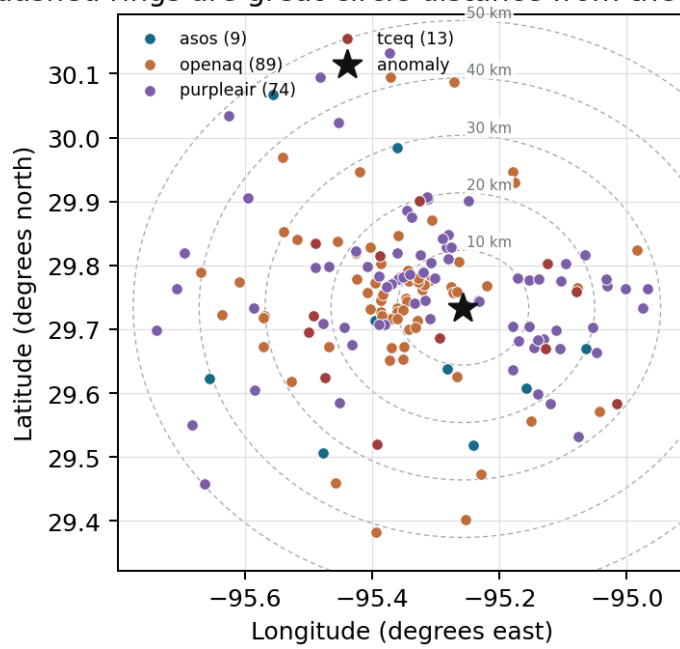
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

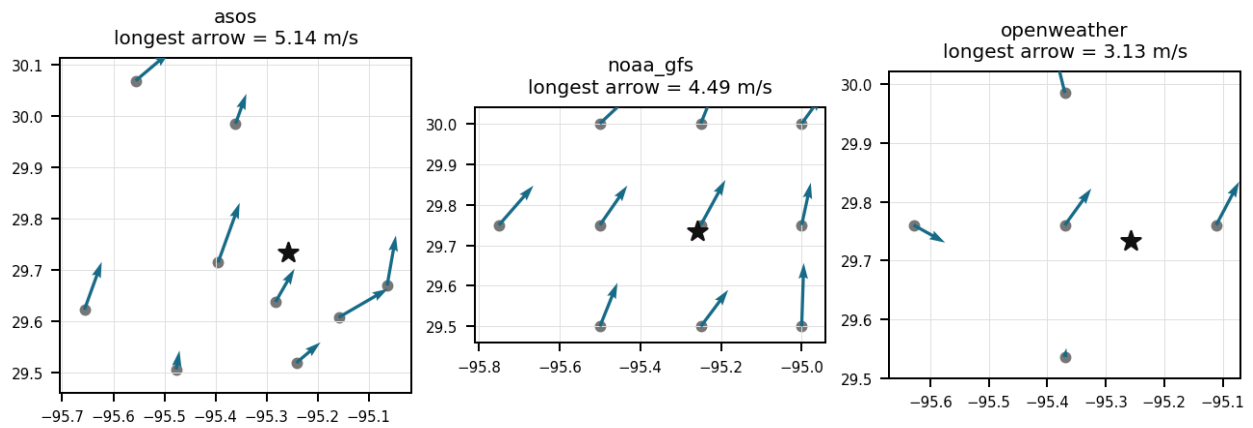
Source	Metric	Values
asos	Relative humidity (humidity)	HOU (10.965 km) 72.88, MCJ (13.448 km) 71.32, EFD (16.998 km) 73.53, T41 (20.02 km) 74.01, LVJ (23.937 km) 72.78, IAH (29.593 km) 68.4, AXH (33.015 km) 73.07, SGR (40.48 km) 75.97, +1 more
asos	Air temperature (temperature)	HOU (10.965 km) 29.85, MCJ (13.448 km) 29.9, EFD (16.998 km) 29.97, T41 (20.02 km) 29.74, LVJ (23.937 km) 29.57, IAH (29.593 km) 30.57, AXH (33.015 km) 28.94, SGR (40.48 km) 29.5, +1 more
asos	Wind direction (wind_direction)	HOU (10.965 km) 188.1, MCJ (13.448 km) 182.1, EFD (16.998 km) 173, T41 (20.02 km) 191.7, LVJ (23.937 km) 131.5, IAH (29.593 km) 179.8, AXH (33.015 km) 103.4, SGR (40.48 km) 173, +1 more
asos	Wind speed (wind_speed)	HOU (10.965 km) 3.64, MCJ (13.448 km) 3.4, EFD (16.998 km) 3.923, T41 (20.02 km) 3.695, LVJ (23.937 km) 2.111, IAH (29.593 km) 2.654, AXH (33.015 km) 1.209, SGR (40.48 km) 3.426, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (1.952 km) 5928, gfs:29.75,-95.50 (23.473 km) 5928, gfs:29.75,-95.00 (24.937 km) 5927, gfs:29.50,-95.25 (26.0 km) 5927, gfs:30.00,-95.25 (29.617 km) 5929, gfs:29.50,-95.50 (34.993 km) 5927, gfs:29.50,-95.00 (35.994 km) 5926, gfs:30.00,-95.50 (37.722 km) 5929, +2 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (1.952 km) 966.3, gfs:29.75,-95.50 (23.473 km) 1053, gfs:29.75,-95.00 (24.937 km) 871.5, gfs:29.50,-95.25 (26.0 km) 680.2, gfs:30.00,-95.25 (29.617 km) 1043, gfs:29.50,-95.50 (34.993 km) 778.2, gfs:29.50,-95.00 (35.994 km) 713.8, gfs:30.00,-95.50 (37.722 km) 1165, +2 more

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (1.952 km) 40.44, gfs:29.75,-95.50 (23.473 km) 40.19, gfs:29.75,-95.00 (24.937 km) 40.92, gfs:29.50,-95.25 (26.0 km) 40.43, gfs:30.00,-95.25 (29.617 km) 40.72, gfs:29.50,-95.50 (34.993 km) 40.16, gfs:29.50,-95.00 (35.994 km) 40.51, gfs:30.00,-95.50 (37.722 km) 40.09, +2 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (1.952 km) 1011, gfs:29.75,-95.50 (23.473 km) 1009, gfs:29.75,-95.00 (24.937 km) 1011, gfs:29.50,-95.25 (26.0 km) 1011, gfs:30.00,-95.25 (29.617 km) 1009, gfs:29.50,-95.50 (34.993 km) 1010, gfs:29.50,-95.00 (35.994 km) 1012, gfs:30.00,-95.50 (37.722 km) 1007, +2 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (1.952 km) 20.42, gfs:29.75,-95.50 (23.473 km) 20.49, gfs:29.75,-95.00 (24.937 km) 20.38, gfs:29.50,-95.25 (26.0 km) 20.43, gfs:30.00,-95.25 (29.617 km) 20.56, gfs:29.50,-95.50 (34.993 km) 20.4, gfs:29.50,-95.00 (35.994 km) 20.53, gfs:30.00,-95.50 (37.722 km) 20.65, +2 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (1.952 km) 0.8703, gfs:29.75,-95.50 (23.473 km) 0.6987, gfs:29.75,-95.00 (24.937 km) 0.4503, gfs:29.50,-95.25 (26.0 km) 0.7012, gfs:30.00,-95.25 (29.617 km) 0.1337, gfs:29.50,-95.50 (34.993 km) 0.202, gfs:29.50,-95.00 (35.994 km) 0.7345, gfs:30.00,-95.50 (37.722 km) 1.551, +2 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (1.952 km) 3.328, gfs:29.75,-95.50 (23.473 km) 3.326, gfs:29.75,-95.00 (24.937 km) 3.057, gfs:29.50,-95.25 (26.0 km) 3.038, gfs:30.00,-95.25 (29.617 km) 2.617, gfs:29.50,-95.50 (34.993 km) 3.421, gfs:29.50,-95.00 (35.994 km) 3.503, gfs:30.00,-95.50 (37.722 km) 2.277, +2 more
openaq	Ozone (ozone)	3378 (0.021 km) 0.01381, 2039 (5.231 km) 0.01818, 325 (6.369 km) 0.005016, 2046 (10.803 km) 0.01553, 320 (12.059 km) 0.01519, 1319333 (14.033 km) 0.02263, 332 (14.34 km) 0.02093, 3214 (14.86 km) 0.02054, +9 more
openaq	PM10 (pm10)	13380082 (0.021 km) 68.4, 15852419 (7.017 km) 25.43, 271 (14.34 km) 30.42
openaq	PM2.5 (pm25)	15539682 (0.005 km) 8.398, 3379 (0.021 km) 12.61, 8967123 (0.058 km) 9.79, 8967479 (0.058 km) 9.324, 9489522 (2.89 km) 6.762, 8967065 (2.933 km) 7.281, 10252270 (4.015 km) 8.045, 14140751 (6.379 km) 7.068, +61 more
openweather	Cloud cover (cloud_cover)	grid:center (11.237 km) 39.88, grid:east (14.451 km) 38.6, grid:south (24.535 km) 34.11, grid:north (29.962 km) 48.18, grid:west (35.93 km) 38
openweather	Relative humidity (humidity)	grid:center (11.237 km) 73.81, grid:east (14.451 km) 78.56, grid:south (24.535 km) 74.86, grid:north (29.962 km) 72.1, grid:west (35.93 km) 74.24
openweather	Precipitation rate (precipitation)	grid:center (11.237 km) 0, grid:east (14.451 km) 0, grid:south (24.535 km) 0, grid:north (29.962 km) 0, grid:west (35.93 km) 0
openweather	Surface pressure (pressure)	grid:center (11.237 km) 1012, grid:east (14.451 km) 1012, grid:south (24.535 km) 1012, grid:north (29.962 km) 1012, grid:west (35.93 km) 1012
openweather	Air temperature (temperature)	grid:center (11.237 km) 30.91, grid:east (14.451 km) 30.19, grid:south (24.535 km) 30.13, grid:north (29.962 km) 30.73, grid:west (35.93 km) 30.35
openweather	Wind direction (wind_direction)	grid:center (11.237 km) 204.7, grid:east (14.451 km) 197.1, grid:south (24.535 km) 197.4, grid:north (29.962 km) 170.4, grid:west (35.93 km) 191.5
openweather	Wind speed (wind_speed)	grid:center (11.237 km) 2.248, grid:east (14.451 km) 3.615, grid:south (24.535 km) 1.605, grid:north (29.962 km) 2.078, grid:west (35.93 km) 1.601
purpleair	PM2.5 (pm25)	99797 (0.6 km) 7.337, 166435 (2.677 km) 10.35, 97279 (2.684 km) 9.325, 6752 (5.276 km) 5.8, 310224 (5.838 km) 9.2, 222601 (6.699 km) 4.718, 133994 (7.377 km) 0.01111, 235425 (8.237 km) 7.336, +66 more
tceq	Carbon monoxide (co)	482011035 (0.0 km) 0.1548, 482011039 (14.355 km) 0.1156, 482011052 (15.44 km) 0.3532
tceq	Nitrogen dioxide (no2)	482011035 (0.0 km) 8.636, 482010416 (6.378 km) 4.567, 482011039 (14.355 km) 2.657, 482010026 (14.878 km) 5.762, 482011052 (15.44 km) 11.86, 482011015 (17.437 km) 4.636, 482010024 (19.743 km) 0.3697, 482011066 (22.737 km) 2.608, +4 more
tceq	Sulfur dioxide (so2)	482011035 (0.0 km) -0.09048, 482011039 (14.355 km) 0.1438, 482010051 (24.236 km) -0.3095

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

The TCEQ monitor 482011035 recorded a carbon monoxide concentration of 0.5 ppm at 2026-07-29T17:00:00+00:00, which is a severe positive anomaly relative to the expected value of about 0.137 ppm at that site and time.

Mark exactly one:

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Note (optional):

Claim 2 of 36

Heavy urban mobile source emissions from surrounding traffic corridors accumulated locally and were transported toward the monitor by light southerly surface winds reported by ASOS and OpenWeather.

Mark exactly one:

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Note (optional):

Claim 3 of 36

Southerly to south-southwesterly near-surface winds of roughly 2 to 3 m/s at the anomaly time could have transported a localized combustion plume from sources south of the monitor toward the site without producing a broad regional pollution episode.

Mark exactly one:

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Note (optional):

Claim 4 of 36

Surface meteorology near the anomaly featured southerly to south-southwesterly winds of about 2 to 2.4 m/s and a deep afternoon planetary boundary layer near 2189 m, which supports transport of a nearby plume to the monitor but does not support strong regional stagnation or shallow-trapping accumulation.

Mark exactly one:

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Note (optional):

Claim 5 of 36

TCEQ surface monitors and OpenAQ sensors recorded simultaneous elevations in carbon monoxide reaching 0.5 ppm, PM10 reaching 120 ug/m3, and NO2 reaching 10.5 ppb at 2026-07-29T17:00:00+00:00, supporting a localized industrial or combustion emission source.

Mark exactly one:

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Note (optional):

Claim 6 of 36

The temperature was unusually high on July 29th, which could have contributed to the formation of ground-level ozone.

Mark exactly one:

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Note (optional):

Claim 7 of 36

Sentinel-5P satellite observations recorded standard baseline total CO column values of approximately 0.0256 mol/m² over the area, indicating that the TCEQ carbon monoxide spike was isolated to the surface boundary layer rather than a widespread regional plume.

Mark exactly one:

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Note (optional):

Claim 8 of 36

Localized industrial combustion or flaring events in eastern Houston released primary pollutants, causing concurrent elevated levels of carbon monoxide, nitrogen dioxide, and particulate matter recorded by TCEQ and OpenAQ sensors.

Mark exactly one:

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Note (optional):

Claim 9 of 36

The anomaly occurred on July 29th at around 17:01:54 UTC, with the nearest sensor located approximately 14.451 km away from the anomaly.

Mark exactly one:

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Note (optional):

Claim 10 of 36

Light winds and stable surface microscale conditions prior to full planetary boundary layer expansion contributed to the localized accumulation of primary combustion emissions, as modeled in GFS data.

Mark exactly one:

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Note (optional):

Claim 11 of 36

The concentration of PM2.5 is elevated to 8.3 ug/m3, which is above the mean value of 8.0665 ug/m3.

Mark exactly one:

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Note (optional):

Claim 12 of 36

The pollutant showing the anomaly is carbon monoxide measured by tceq.

Mark exactly one:

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Note (optional):

Claim 13 of 36

The air-quality anomaly in Houston, Texas on July 29th was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 14 of 36

The tceq carbon monoxide anomaly is consistent with a localized primary combustion plume because the anomalous monitor measured 0.5 ppm at 17:00 UTC while other nearby tceq carbon monoxide monitors within about 15 km had lower period means of 0.1156 to 0.3532 ppm and the coincident tceq nitrogen dioxide value at the same monitor was elevated at 10.5 ppb, which is a typical co-emitted combustion tracer.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 15 of 36

Sentinel-5P satellite retrievals showed a stable regional CO column density near 0.0265 mol/m² on July 29, 2026, indicating the anomaly was limited to the surface layer.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 16 of 36

The air quality in Houston, Texas is poor due to high levels of SO₂.

Mark exactly one:

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Note (optional):

Claim 17 of 36

NO₂ levels were also elevated in Houston, Texas on July 29th, suggesting increased emissions from industrial or vehicular sources.

Mark exactly one:

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Note (optional):

Claim 18 of 36

OpenAQ sensors co-located near the anomaly reported elevated PM₁₀ concentrations reaching 120.0 ug/m³ at 17:00 UTC on July 29, 2026.

Mark exactly one:

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Note (optional):

Claim 19 of 36

A severe carbon monoxide concentration of 0.5 ppm was recorded at coordinates (29.734, -95.258) on July 29, 2026, at 17:00 UTC, significantly exceeding the expected baseline value of 0.137 ppm.

Mark exactly one:

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Note (optional):

Claim 20 of 36

Nitrogen dioxide concentration measured 10.5 ppb at the primary anomaly monitor location on July 29, 2026, at 17:00 UTC.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 21 of 36

Meteorology from asos, openweather, and gfs supports transport of a localized plume rather than strong stagnation because near the anomaly winds were southerly to south-southwesterly at about 190 to 208 degrees with speeds around 2.1 to 2.4 m/s and the afternoon boundary layer was deep at about 2189 m, allowing advection from upwind source areas while not favoring citywide surface accumulation.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 22 of 36

A region-wide CO accumulation event or wildfire-smoke-driven anomaly is less plausible because daytime boundary-layer depth was high, no precipitation-related stagnation signal was present, satellite CO column was not elevated over the area, and surface PM2.5 remained only modestly elevated even though one nearby PM10 sensor was high.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 23 of 36

TCEQ monitor 482011035 recorded a carbon monoxide concentration of 0.5 ppm alongside an NO2 level of 10.5 ppb at 17:00 UTC on July 29, 2026.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 24 of 36

The data contradict a regional smoke or widespread secondary-pollution explanation because sentinel5p showed a relatively low local carbon monoxide column on 2026-07-29 compared with the adjacent days, ozone near the monitor was only 0.018 ppm, and purpleair and openaq fine-particle levels were modest near the event even though one collocated openaq PM10 sensor was high, indicating a near-source coarse or mixed local plume rather than regional smoke.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 25 of 36

The air quality in Houston, Texas is poor due to high levels of NO2.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 26 of 36

The air quality in Houston, Texas is poor due to high levels of PM2.5.

Mark exactly one:

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Note (optional):

Claim 27 of 36

The tceq carbon monoxide measurement at coordinates (29.734, -95.258) was 0.5 ppm at 2026-07-29T17:00:00+00:00.

Mark exactly one:

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Note (optional):

Claim 28 of 36

A short-lived local combustion source near the Houston monitor, such as industrial process emissions, flaring-related combustion, heavy-duty traffic, or nonroad equipment, is the most plausible physical cause of the severe CO spike because the CO enhancement was large at one TCEQ site while co-pollutants remained only modestly elevated or low at that location.

Mark exactly one:

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Note (optional):

Claim 29 of 36

The pollutant responsible for the anomaly is PM2.5.

Mark exactly one:

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Note (optional):

Claim 30 of 36

The wind direction and speed on July 29th were conducive to the accumulation of pollutants in the area, leading to the observed air-quality anomaly.

Mark exactly one:

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Note (optional):

Claim 31 of 36

ASOS surface stations and OpenWeather data reported light south-southwesterly winds around 2.0 m/s during the anomaly, providing weak atmospheric dispersion that allowed localized primary pollutants to accumulate near ground level.

Mark exactly one:

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Note (optional):

Claim 32 of 36

The PM2.5 levels exceeded the normal range in Houston, Texas on July 29th, indicating a significant increase in particulate matter in the air.

Mark exactly one:

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Note (optional):

Claim 33 of 36

Coinciding with the carbon monoxide spike on July 29, 2026, at 17:00 UTC, PM10 and PM2.5 concentrations reached 120.0 ug/m3 and 15.7 ug/m3 respectively at a location 0.021 km from the anomaly.

Mark exactly one:

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Note (optional):

Claim 34 of 36

Regional observations do not support a broad wildfire-smoke or region-wide CO buildup event because Sentinel-5P CO column near Houston was not elevated on 2026-07-29, while the anomalous site simultaneously showed elevated NO2 and high local PM10 consistent with a localized fresh combustion plume.

Mark exactly one:

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Note (optional):

Claim 35 of 36

The tceq carbon monoxide measurement of 0.5 ppm at 2026-07-29T17:00:00+00:00 exceeded the expected value of 0.13695652173913042 ppm with a z-score of 6.145967130068515, and the anomaly was labeled severe.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 36 of 36

A nearby industrial facility may have released excessive amounts of sulfur dioxide and nitrogen oxides into the atmosphere, causing the air-quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
